

POLYCYSTIC OVARY SYNDROME(PCOS) AND MACHINE LEARNING

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Abstract

PCOS (Polycystic Ovary Syndrome) is an endocrine disorder which affects women who fall in the fertility age bracket and inflicts harm on their metabolic and reproductive health. Despite early diagnosis being pivotal for effective treatments, the traditional diagnostic methods suffer from shortcomings like high costs and case specific symptoms. Machine learning and deep learning offer promising prospects for the prediction of PCOS as they offer ground for higher accuracy and efficiency. This review paper aims to consolidate existing research on prediction of PCOS using ML/DL, to analyze the performance of various algorithms, to identify the gaps in current methodologies and to propose future research prospects [4][5]. The findings of this paper will contribute to the advancement of automated detection/diagnostic tools for PCOS, will improve the chances of early detection of PCOS and ease the creation of personalized treatment strategies.

Keywords: Polycystic Ovary Syndrome (PCOS), Machine Learning, Deep Learning, Early Diagnosis, Automated Detection, Personalized Treatment

1. Introduction

PCOS (Polycystic ovary syndrome) is a common hormonal condition that affects women of reproductive age [6]. Women who have high levels of androgens, which can manifest as excess testosterone or androgen disorders such as hyperandrogenemia are at high risk [7]. Majority of the population which suffers from PCOS has been observed to develop insulin resistance, which in turn results in increased secretion of androgen [8].

PCOS usually starts during adolescence, although the symptoms will change over time. PCOS affects an estimated 6–13% of women of reproductive age [9]. Statistics show that up to 70% of affected women remain undiagnosed worldwide [10]. PCOS often results in hormonal imbalances and irregular menstrual cycles and excess androgen levels, in some cases, it might also cause cysts in the ovaries, type 2 diabetes, and cardiovascular diseases [11][12]. Irregular periods with a lack of ovulation can make it difficult to become pregnant, making PCOS a leading cause of infertility. Additionally, PCOS is associated with an increased risk of breast cancer [13]. As a result, the treatment of PCOS has become vital and there is a rising need for methods of timely detection and treatment which are also scalable and accessible [14][15].

Traditionally used diagnostic techniques depend on clinical, biochemical, and imaging criteria—may often be imprecise, demand a high amount of resources and are dependent on specialized expertise [16]. The application of machine learning and deep learning represents a critical step forward in addressing these diagnostic shortcomings [1]. Models for prediction of PCOS which use varied datasets—including clinical records, hormonal profiles, and imaging data—are becoming a possibility with recent advances in ML and DL as they allow us to identify complex patterns in diverse data sets [2][5]. Studies making use of algorithms like Random Forest, Support Vector Machines, Artificial Neural Networks, and other ensemble methods have scope for providing promising accuracies, as high as above 95% [1][4][5]. However, there are still many unresolved issues pertaining to selection of features, dataset imbalance, and model interpretability [3].

Certain genetic markers for PCOS can serve as predictive and diagnostic tools, but their effectiveness may be limited by the condition's intricate genetic nature [17]. The present situation of the use of Machine learning and deep learning in the prediction of PCOS will be assessed in this review, placing significance in strengths, constraints, and future research prospects [4][5]. This review also discusses explainable AI techniques to enhance model transparency to foster clinician trust, highlights the need for diverse datasets and discusses integration in the healthcare system [3][5].

2. Literature Review

In their paper Zad et al. (2024)[1] learned machine learning models for predicting polycystic ovary syndrome (PCOS) from 30,601 women's aged 18-45 years old electronic health records (EHR) at Boston Medical Center by using Rotterdam criteria and merging ICD-9 codes, labs, and ultrasonography reports. Logistic regression, support vector machines, gradient boosted trees, and random forests were used in the study, with four models of diagnosed and undiagnosed PCOS cases with AUCs(Area under the curve) of 85%, 81%, 80%, and 82%. Strong predictors were hormone values (using a multilayer perceptron score), obesity, and gravidity, with the latter inversely related. The strength of the study is having a high-volume, high-diversity sample from a SafetyNet hospital, rather than specifically fertility clinics, and having three data streams to give maximal predictive power. Its hospital-based EHR data limitation confines generalizability to non-healthcare-interacting or rural populations, and omitting variables such as AMH, diet, or genetics might be missing other predictors. In larger populations, validation will tell us how valuable it is.

Elmannai et al. (2023) [3] designed a machine learning algorithm for early polycystic ovary syndrome (PCOS) diagnosis from a Kaggle dataset of 541 instances and 41 features, dealing with class imbalance and feature selection optimization using mutual information, recursive feature elimination (RFE), and tree-based methods. The paper contrasted different models, such as logistic regression, random forest, and random support vector machines, but suggested a stacking ensemble model of high-performance base learners (e.g., random forest) augmented with a random forest meta-learner, tuned with Bayesian optimization. The RFE stacking model performed best with 100% accuracy, precision, recall, and F1-score on an 80:20 train-test split, better than existing work (e.g., 98% accuracy in multi-stacking models). Highlighting the focus on explainable AI, authors presented global and local explanations through SHAP values and visualization in the forms of waterfall and force plots for model truthfulness. Optimal performance metrics though are alarming on the risk for

overfitting and the singular dependency of research on one data set for reduced generalizability..

Nasim et al. (2022) [4] created a machine learning model for the prediction of polycystic ovary syndrome (PCOS), employing multi-clinic data gathered from Kerala, India, on 41 clinical and physical features from a set of 541 patients. The paper introduced a new feature selection algorithm, CS-PCOS, based on optimum chi-squared techniques, which chose the 20 most significant such as prolactin, blood pressure (systolic and diastolic), thyroid-stimulating hormone, respiratory rate, and pregnancy status. Ten parameter-tuned machine learning models were tried out, and the optimal performance, satisfactory performance measures (100% accuracy, precision, recall, and F1-score), and training time of 0.002 seconds produced by the Gaussian Naive Bayes (GNB) model on a train-test ratio of 80:20 were observed. Model robustness was also confirmed through 10-fold cross-validation with 100% accuracy. They have also performed deep exploratory data analysis (PEDA) to discover patterns concerning PCOS like correlation and feature distribution. Compared to the past works reported in the literature (e.g., 91% accuracy using RFLR), CS-PCOS+GNB method outperforms current state of art methods. But ideal metrics suggest possible overfitting and generalizability constraints since a single-small dataset is employed in the research; future research will include larger datasets and deep-learning methods.

Mridul et al. (2024) [6] explored PCOS prediction on a large dataset of conventional and deep learning models on a 540 patient dataset with 46 clinical features, such as hormonal and metabolic markers, from Kaggle. The initially unbalanced dataset (363 non-PCOS and 177 PCOS) was resampled using ADASYN and divided as an 80:20 train-test set. The research applied 16 classifiers (Logistic Regression (LR), Random Forest (RF), Support Vector Classifier (SVC), Artificial Neural Network (ANN), and Recurrent Neural Network (RNN) and ensemble models like bagging, boosting, stacking, and voting. Optimal performance was achieved with the use of hyperparameter tuning and Boosted RF and SVC delivered 99.32% accuracy, precision, specificity, and F1-score, while Stacking Classifier RDAS had a 99.32% accuracy. Deep learning algorithms like ANN (98.63% accuracy) outperformed others like CNN (89.04%) at picking up on intricate patterns. Feature importance was examined but not single selected features. High accuracy in the study implies high predictive capacity, although a single dataset used and susceptibility to overfitting need to be tested with varied and large datasets. Future work will incorporate multi-modal data (e.g., imaging, genetic) and create explainable AI models for clinical deployment.

Khanna et al. (2023) [7] created an explainable machine learning pipeline for diagnosing polycystic ovary syndrome (PCOS) based on a Kaggle dataset of 541 fertile women from Kerala, India, with 43 clinical features (24 non-invasive, 19 invasive) and a target variable of PCOS diagnosis (177 PCOS vs. 364 non-PCOS cases). Pre-processing was performed on the dataset with median/mode imputation of missing values and balancing. Three feature selection algorithms—Harris Hawks Optimization (HHO), Salp Swarm Optimization (SSA), and Mutual Information (MI)—selected 15–19 important features, including follicle numbers, weight gain, and hormonal markers. Twelve machine learning classifiers (e.g., Logistic Regression, Random forest) and two deep learning models were evaluated, with a multi-level stacking ensemble (STACK-3) trained on MI-engineered data yielding the best performance: 98% accuracy, 97% precision, 98% recall, and 98% F1-score. Explainable AI

(XAI) techniques, including SHAP, LIME, Qlattice, and Random Forest feature importance, were applied to tree-based models, which reported follicle numbers, hair growth, and skin darkening as important predictors. Compared with deep learning models (DNN: 94% accuracy; 1D-CNN: 90% accuracy), the stacking ensemble had superior performance, though its complexity may limit clinical implementation. Future work is recommended in real-time screening interfaces and non-invasive feature-based applications, with the need for larger, diverse datasets to enhance generalizability.

Study	Dataset	Feature selection	Algorithms	Strengths	Limitations
Zad et al.(2024)	Kaggle,541 patients,41 features,177 PCOS	Pearson correlation(10 features:follicle nos.,cycle irregularity	11 ML(LR,RF,SV M,XGBoost)	Non-Invasive focus,high accuracy	Single dataset,limited generalizability
Elmannai et al.(2023)	Kaggle,541 patients,41 features,177 PCOS	Unspecified(10 features:follicle counts,hormonal ratios)	9 ML(DT,SVM,R F,XGBoost)	Gradient Boosting:94.44% accuracy	Small dataset,lacks diversity
Nasim et al.(2022)	IoT,150 patients,45 features	RFE(BMI,cycle length,hormones)	7 ML(KNN,SVM, RF,Extra Trees)	Extra Trees:96.67% accuracy	Small sample,no external validation
Mridul et al.(2023)	Kaggle,541 patients,46 features,177 PCOS	Feature importance (unspecified features)	16 ML/DL(RF,SV C,ANN,RNN)	Boosted RF/SVC/Stacking :93.32% accuracy	Overfitting risk,single dataset
Khanna et al.(2023)	Kaggle,541 patients,43 features,177 PCOS	HHO,SSA,MI(15-19 features: follicle nos., weight gain)	12 ML,2 DL(DNN,1DC NN)	Stacking (STACK-3):98% accuracy	Complex model,limited scalability

TABLE 1. Summarizing key points

3. Methodology

3.1. Research Questions

To guide the systematic review, the following research questions (RQs) were formulated:

- **RQ1:** What keywords are prevalent in machine learning research for PCOS detection?
- **RQ2:** Which machine learning algorithms are most frequently used for PCOS identification?
- **RQ3:** What journals have published studies on ML-based PCOS detection?
- **RQ4:** Which algorithm demonstrates the highest performance in PCOS analysis?
- **RQ5:** How many studies focus on PCOS detection using machine learning?

3.2. Literature Search Strategy

- **Databases:**
Scopus, PubMed, Web of Science and IEEE Xplore were searched to ensure comprehensive coverage of peer-reviewed journal articles.
- **Keywords:**
Boolean search strings combined PCOS-related terms with ML and explainability techniques, emphasizing SHAP:
 - ("Polycystic Ovarian Syndrome" OR "PCOS") AND "Machine Learning"
 - ("PCOS" OR "Polycystic Ovary Syndrome") AND ("Detection" OR "Identification")
 - ("PCOS" AND "SHAP" OR "Explainable AI")
 - ("PCOS" AND ("SVM" OR "Random Forest" OR "Neural Network"))
- **Filters:**
 - Publication years: 2020–2025.
 - Language: English.
 - Document type: Peer-reviewed journal articles only (excluding conference papers, reviews, or short articles).
- **Search Tools:**
Web of Science and ABCD Index was used to filter Scopus-indexed articles. The initial search yielded 200–400 articles, but only 35 were eligible for this research after applying quality filters.

3.3. Quality Evaluation

3.3.1. Inclusion and Exclusion Criteria

In order to obtain articles relevant to a research topic, specific criteria such as inclusion and exclusion criteria must be chosen. Entry criteria are used as criteria for document selection, and exit criteria are not used in document selection (refer to Table 1).

S. No.	Inclusion Criteria	Exclusion Criteria
1	Published between 2020-2025	Studies outside the data range or unpublished
2	English-language journal articles	Non-English publications
3	Indexed in SCIE, SSCI, ESCI, or Scopus with SJR ≥ 0.15	Non-indexed or low-SJR journals
4	Focus on ML for PCOS detection, including SHAP/explainability	Studies without ML or unrelated to PCOS
5	Peer-reviewed and full-text available	Abstracts-only, conference papers, or non-peer-reviewed sources

TABLE 2. INCLUSION-EXCLUSION CRITERIA TABLE

Once the inclusion criteria were set, the study commenced with the search for relevant articles using the PRISMA (Preferred Reporting Items for Systematic Review and Meta-analysis) protocol. With PRISMA, there are 3 steps to remove research reviews to make them appropriate and aligned with the research topic. These steps are described in the following flowchart (FIGURE 1).

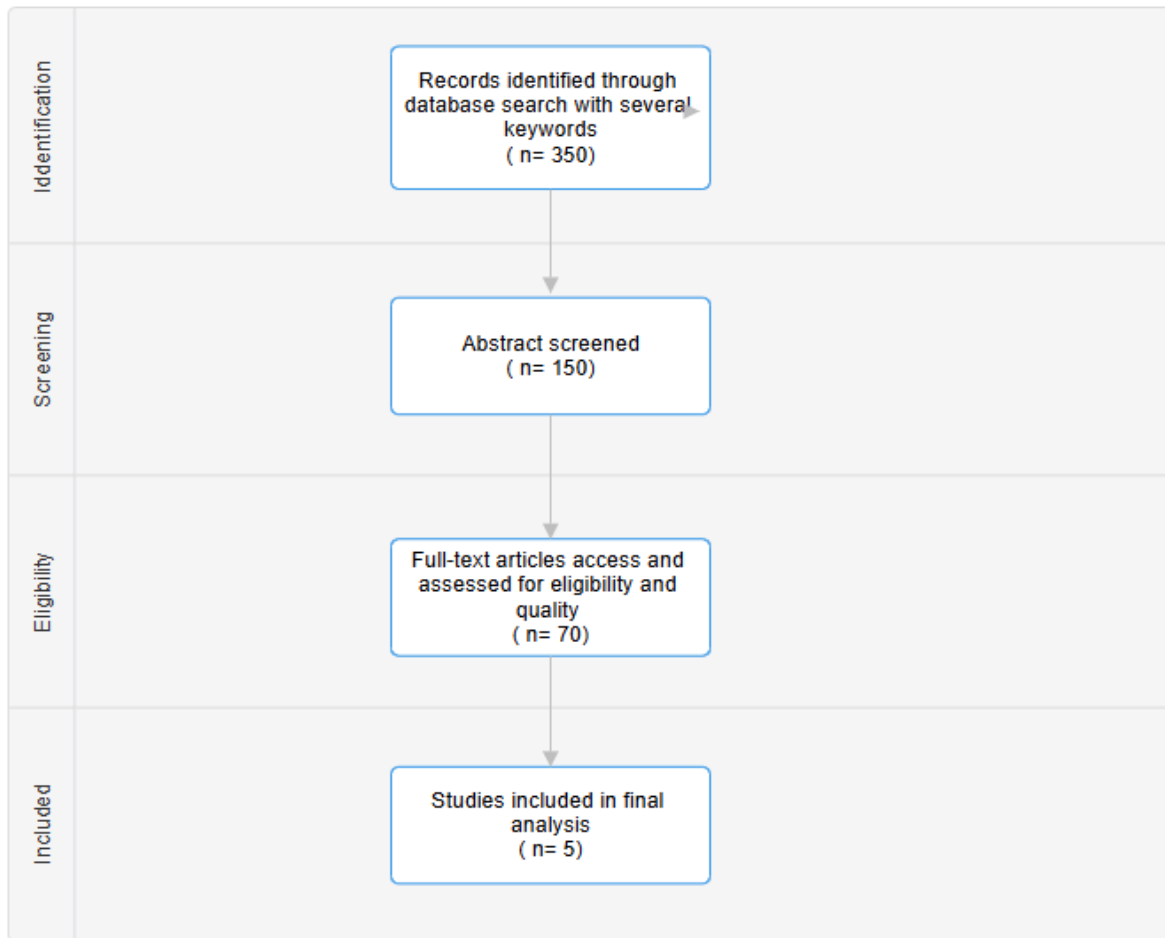


FIGURE 1. PRISMA FLOW CHART

3.3.2. Quality Metrics

- **Tools:** ABCD Index website for Scopus metrics and Web of Science for SCIE/SSCI/ESCI indexing verification.
- **Metrics:** SJR score (≥ 0.15), SCIE/SSCI/ESCI/Scopus indexing, quartile ranking, and citation count.
- **Outcome:** Only journal articles meeting all inclusion criteria, with a focus on ML and SHAP/explainability for PCOS detection, were analyzed

3.4. Using Machine Learning to Identify PCOS

Recent developments indicate that Machine Learning(ML) came out as a very useful technique for PCOS(Polycystic Ovary Syndrome) early identification. ML models provide the ability to analyze large, complex clinical datasets and discover delicate patterns which traditional diagnostic methods might fail to notice [1][2]. Out of the studies reviewed, various machine learning algorithms have been used, including Logistic Regression (LR), Random Forest (RF), Support Vector Machines (SVM), Gaussian Naive Bayes (GNB), Decision Trees (DT), and many deep learning architectures like Multilayer Perceptrons (MLPs) and convolutional neural networks(CNNs) [1][4][5]. Researchers have focused more on feature

selection methods in order to improve model performance [2][3]. Techniques such as optimized chi-square selection (CS-PCOS), univariate feature selection, and Recursive Feature Elimination (RFE) were applied to identify clinically relevant predictors [2][18]. By selecting an optimized set of features, models reduced overfitting and improved both accuracy and computational efficiency [3][4].

Moreover, advanced methods like explainable artificial intelligence (XAI) were included into the model design to improve the understanding and trust. Elmannai et al., (2023) introduced local and global concepts for explainability assuring that both clinicians and patients could understand the predictions of the model [3]. Deep Learning models, particularly the use of multilayer perceptrons (MLP), increased the power to capture non-linear relationships among hormonal signals such as FSH, LH, SHBG, and estradiol [4].

Overall, these ML-driven approaches achieved exceptional results, with some models, such as the Gaussian Naive Bayes with CS-PCOS feature, reaching 100% accuracy [2]. The collective proof highlights that machine learning when combined with careful feature engineering and explainability frameworks, holds a great potential in transforming PCOS diagnosis and making timely clinical measures easier.

3.5. Explainable AI (XAI) like SHAP for PCOS Models

In clinical settings, interpretability of a model is crucial for doctors and experts to build trust in AI-driven models, especially in complex conditions like Polycystic Ovary Syndrome (PCOS) [5]. Doctors and researchers need to understand why a model makes certain predictions to ensure correct predictions, figure out the root of the problem and take effective actions [19]. For instance, knowing which features (e.g., insulin resistance or Anti-Müllerian Hormone (AMH) levels) contribute most to a PCOS diagnosis helps clinicians prioritize certain factors while conducting treatments [5]. Black-box models, despite high accuracy, often fail to provide this transparency, making Explainable AI (XAI) techniques like SHAP essential [9].

SHAP values, rooted in cooperative game theory, quantify the contribution of each feature to a model's prediction by fairly distributing the "payout" (prediction output) among input features [9]. Unlike simpler methods like feature importance plots which only show global relevance or LIME (Local Interpretable Model-agnostic Explanations) which approximates locally, SHAP provides consistent and accurate attributions at both global and local levels [9][19]. For example, SHAP can reveal that high insulin levels disproportionately increase PCOS risk in certain patient subgroups, while AMH is more influential in others [5].

SHAP analysis helps doctors understand how different health factors contribute to PCOS diagnosis (refer to Figure 2 and Figure 3). For example, research using SHAP has shown that insulin resistance is typically the strongest indicator of PCOS overall [5]. But it also reveals more complex patterns - like how having both high testosterone and high BMI together creates a much greater risk than either factor alone [20]. The real power of SHAP comes from its visual explanations that make these technical findings easy to understand [9]. Doctors can see exactly why the model flagged certain patients as high-risk, even when their symptoms appear borderline [5]. This makes AI predictions more trustworthy and useful in real clinical practice, combining computer accuracy with human understanding [19].

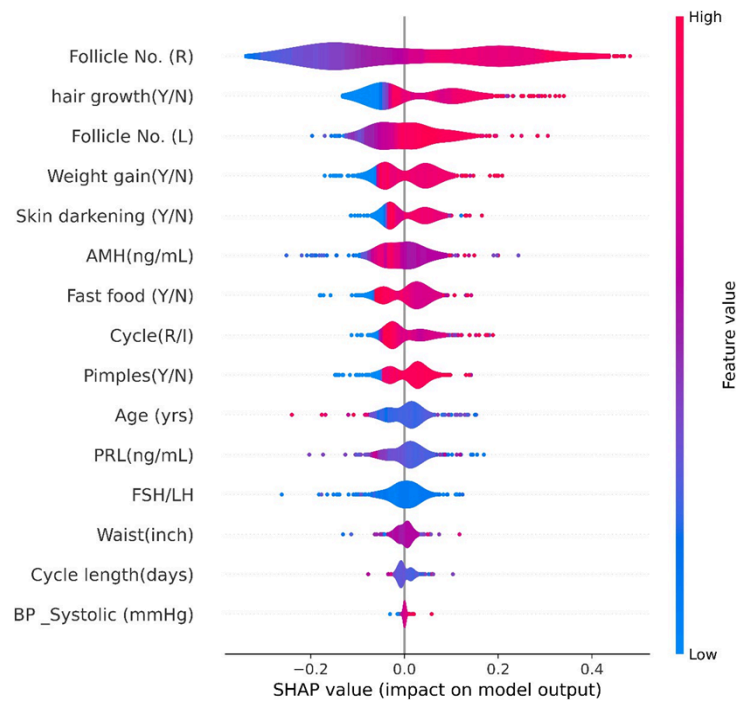


FIGURE 2. SHAP Violin Plot

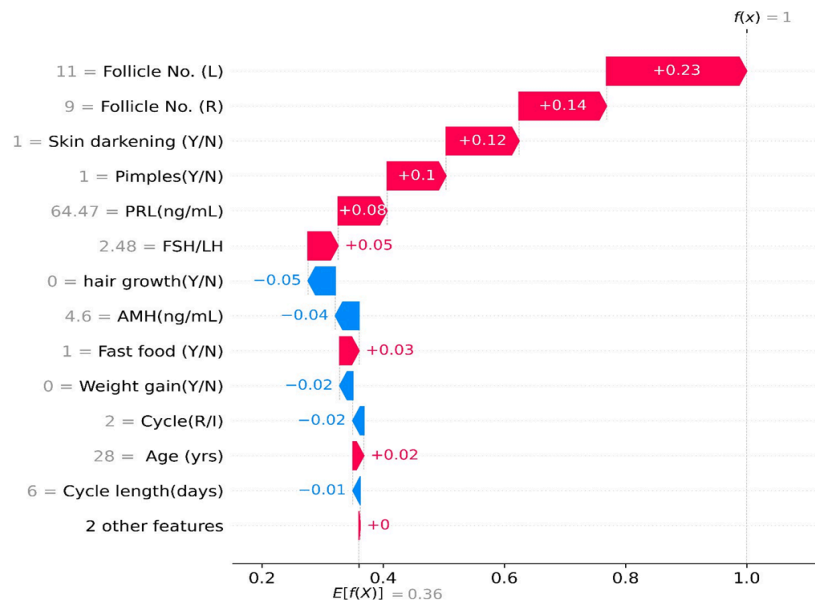


Figure 3. SHAP Waterfall Plot

Recent research has demonstrated SHAP's utility in PCOS diagnostics [5]. For instance, a 2023 study published in Scientific Reports used SHAP to interpret an ensemble model for PCOS prediction, revealing that fasting insulin and LH/FSH ratio were the most influential features, aligning with known endocrine dysregulation in PCOS [5]. Another study in Journal of Biomedical Informatics (2022) applied SHAP to explain risk stratification in heterogeneous

PCOS subgroups, showing that AMH levels were critical for younger patients, while insulin resistance dominated in older cohorts [21]. These insights not only validate clinical knowledge but also uncover subtle patterns that might be missed with traditional statistical methods [22].

4. Comparative Analysis of Machine Learning Approaches for PCOS Detection

Study	Models Used	Best Performing Model(s)	Accuracy/AUC	Special Techniques
Nasim et al. (2022)	GNB, RF, SVM, SGD, LIR, MLP, KNC, LOR, GBC, BR	Gaussian Naïve Bayes (GNB)	100% (with CS-PCOS)	Optimized Chi-Squared Feature Selection (CS-PCOS), K-Fold Cross Validation
Elmannai et al. (2023)	LR, RF, DT, NB, SVM, KNN, XGBoost, AdaBoost, Stacked Model	Stacked ML with RFE	100%	SMOTE-ENN balancing, Bayesian Optimization, Explainable AI (XAI)
Mridul et al. (2024)	ANN, CNN, RNN, LSTM, BLSTM, RF, SVM, GB, KNN, ABC, DT, GNB, RC, PA	Stacking Classifier (RDAS)	99.32%	Ensemble Learning, Hyperparameter Tuning
Zad et al. (2024)	Logistic Regression, SVM, Gradient Boosted Trees, Random Forest, MLP Neural Network	Logistic Regression and Random Forest	AUC: 80%–85%	Large EHR Dataset (30,601 records), MLP Score for Hormone Mapping

5. Clinical Implications, Limitations and Future Directions

5.1. Clinical implications

Clinical Implications, Limitations and Future Directions 5.1. Clinical implications ML and DL predictive models of PCOS could potentially transform clinical practice by enabling early diagnosis and personalized treatment [4]. These models could potentially predict risk in high-risk patients before the development of full-blown symptoms, in time considering PCOS's association with infertility, type 2 diabetes, cardiovascular disease, and psychiatric disease [11]. Early detection through correct predictions would enable clinicians to treat early, for instance, through lifestyle modification or drug therapy, to avert symptoms and long-term results [23]. Deployment of these models on electronic health record systems would enable clinicians to view real-time risk scores, to guide decision-making and patient selection for further diagnostic assessment or screening [4]. Explanation of model predictions, backed by current standards like the Rotterdam criteria, would enable clinician confidence, to enable deployment in varied healthcare settings [5]. This would prevent delay in diagnosis, particularly in disadvantaged populations, and enable personalized treatment plans, optimizing patient quality of life and minimizing healthcare cost [24]. Successful PCOS prediction models ultimately would enable clinicians to deliver proactive, personalized care, treating current symptoms and future health risk in women worldwide [1].

5.2. Limitations

Interpretability is still a key challenge because complex ML and DL models tend to be "black boxes" to clinicians who require transparent decision-making tools in which they can have trust and act upon predictions [5]. Efforts to integrate Explainable AI (XAI) techniques have been made, but they tend to fail in generating actionable insights in line with clinical diagnostic criteria, e.g., the Rotterdam criteria [25]. This transparency gap diminishes clinical trust and the rate at which these models are implemented in clinical settings [3].

One of the most significant challenges to PCOS prediction is the dependence on small, non-representative datasets [4]. Most existing research relies on datasets with only a few hundred patient records, commonly gathered from certain geographic areas [1]. These datasets are limited in their ability to reflect the full range of PCOS presentations within representative ethnic, socioeconomic, or geographic populations [26]. They also tend to be missing significant data, like genetic markers, rich hormonal profiles, or ultrasound images, which might enhance predictive performance [17]. This narrow scope limits the models' capacity to generalize well, and it becomes challenging to use them in diverse clinical environments globally [8]. Synthetic data generation algorithms are frequently employed by researchers, but these introduce noise and artificial trends in data, leading to spurious predictions [13]. This is an overfitting problem that is useful when models are doing nearly optimally on training sets but fail to generalize to new patients [4]. Model robustness uncertainty is also generated by not validating on independent, diverse datasets since high accuracy can be due to memorization of some data and not actual predictive ability.

The incorporation of ML and DL models within clinical practice is also fraught with

challenges [6]. Most models rely on expert data, like follicle counts or hormone levels, that is not always available in the healthcare environment [7]. In addition, such models are currently not integrated in electronic health records, constraining their application in practice [8]. Another barrier comes from the computational complexity of more advanced models since they demand enormous resources to train and deploy them, rendering them less suitable for environments with constrained resources such as rural clinics or mobile health settings [15]

5.3. Future Directions

For Doctors to accept these tools, AI models need to provide their predictions in simple language with reasoning [3]. Explainable AI and techniques such as SHAP values assist by revealing what inputs (symptoms, hormone levels) drove each decision [9]. The key is translating these explanations into traditional PCOS diagnostic criteria doctors already apply [25].

In the future, these models can also be expanded to model long-term outcomes, such as PCOS-associated cardiovascular diseases or diabetes, which will enhance preventive medicine and personalized medicine [8]. Creating light models along with thorough scalability testing across various healthcare infrastructures will allow these tools to reach more people [4]. Effort should be on user-centered interfaces that align with clinical workflows in order to support healthcare providers [4].

To address challenges, the future should be focused on building large, diverse, and multi-modal datasets [8]. Data collection in collaboration with global healthcare centers, covering diverse populations, data from genetic, hormonal, and imaging tests, along with clear metadata about data collection methods, will increase predictive accuracy and generalizability [17]. Techniques like regularization, k-fold cross-validation, and noisy validation on data should be considered to avoid overfitting [4]. Subgroup analyses by demographic subgroups like age or ethnicity will ensure fairness of prediction, removing bias and increasing equity [26]. Lastly, the creation of collaborative ecosystems is critical to bridge research to practice [4]. Involving researchers, clinicians, industry stakeholders, and patients will ensure the creation of user-centered tools [24]. Maintaining ethical and regulatory standards, for instance by following FDA guidelines, through open reporting of model bias and limitations will ensure trust and clinical uptake [6].

6. Conclusion

The application of machine learning in the detection and prediction of Polycystic Ovary Syndrome (PCOS) indicates a significant progress in the field of medical diagnostics. It is clear that the machine learning models, especially ensemble techniques and optimised feature selection methods can achieve exceptional accuracy, sometimes reaching up to 100%. These models not only improve diagnostic precision but also reduce the time and cost related with traditional clinical investigations. By identifying important clinical and biochemical features through advanced selection techniques such as chi-squared optimization and univariate running, machine learning models facilitate the diagnostic process, offering efficient ways for early detection.

Furthermore, the use of explainable artificial intelligence (XAI) frameworks has increased trust in machine learning models by providing transparency in prediction results which is an important factor for real world medical use. Studies based on large datasets, such as Electronic Health Records (EHRs), have further shown a good ease of use of machine learning approaches on a large scale although they emphasise on the need for further approval from different hospital systems and populations.

In comparative evaluation, ensemble models and stacked performed far better than the conventional single-model methods, being more accurate, more effective at dealing with uneven distribution, and greater understanding. Still, problems of data imbalance, model overfitting, and explainability requirements need to be addressed in order to guarantee the ethical and practical application of such technologies in real-world clinical practices.

Thus, machine learning provides a positive approach for the early detection and treatment of PCOS, which may improve patient outcomes, reduce healthcare expenses, and improve quality of life for women suffering from the condition. In the future, however, research must aim to make models more flexible, test them in practice, and implement them in clinical settings to fully realize the potential of AI in diagnosing PCOS.

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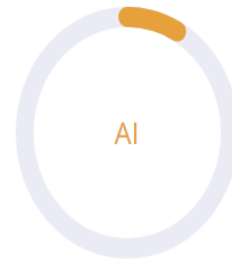
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POLYCYSTIC OVARY SYNDROME(PCOS) AND MACHINE LEARNING Sarthak

Chaturvedi*, Riya Garg*, Simran Pahwa* *Department of Computer Science , Keshav Mahavidyalaya, India Abstract PCOS (Polycystic Ovary Syndrome) is an endocrine disorder which affects women who fall in the fertility age bracket and inflicts harm on their metabolic and reproductive health. Despite early diagnosis being pivotal for effective treatments, the traditional diagnostic methods suffer from shortcomings like high costs and case specific symptoms. Machine learning and deep learning offer promising prospects for the prediction of PCOS as they offer ground for higher accuracy and efficiency. This review paper aims to consolidate existing research on prediction of PCOS using ML/DL, to analyze the performance of various algorithms, to identify the gaps in current methodologies and to propose future research prospects [4][5]. The findings of this paper will contribute to the advancement of automated detection/diagnostic tools for PCOS, will improve the chances of early detection of PCOS and ease the creation of personalized treatment strategies.

Keywords: Polycystic Ovary Syndrome (PCOS), Machine Learning, Deep Learning, Early Diagnosis, Automated Detection, Personalized Treatment 1. Introduction PCOS



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